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St. Joseph's college of Arts & Science (Autonomous)

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B. Sc DEGREE EXAMINATION-APRIL 2021

BC203S-BIOMOLECULES-II

Time: 3 hrs **Max Marks: 75**

SECTION–A (25X1=25)

Answer ALL the questions:

I choose the best answer: (10X1=10)

1. Which one is a fibrous protein?

   a)Actin b)myocin c)albumin d)keratin

2. Alpha helix structure of proteins is stabilized by_____

   a) Peptide bond b) Van der Waals forces

   c) H-bond d)Dipole-dipole interactions.

3. Each polypeptide in a protein has an amino acid sequence is said to be_____

   a)primary structure b) Secondary structure

   c) Tertiary structure d) Quaternary structure

4. Which of the following is a conjugated protein?

   a) Phospho protein b) Glycoprotein c) Chromoprotein d) All of these.

5. Which function group participates in disulphide bond formation in protein?

   a)Thiolactone b)Thiol c)Thioether d)Thioester.

6. A Protein when heated with Conc.Niric acid give yellow colour.This is
   a)Hoppe' s test b)Acid base test c)Biuret' test d)Xanthoprotic test
7. Which amino acid has an imidazole ring?
   a) Alanine b)Histidine c)Leucine d)Tyrosine.
8. Among the following which one is a monounsaturated fatty acid?
   a)Oleic acid b)Palmitic acid c) Malalic acid d) Lenolenic acid.
9. Phopho lipid is ---- type of lipids?
   a) Simple b) Compound c) Derived d) None.
10. Which fatty acid reduces the blood glycerides level
    a) Omega fatty acid b) Alpha fatty acid c)Beta fatty acid d) None

II. Say true or false from the following questions

11. Glycine amino acid is optically inactive
12. Phenyl Alanine is a non essential amino acid.
13. Collagen is a fibrous protein
14. Stearic acid is an essential fatty acid
15. Glycerol and fatty acid, both are basic units of lipids

III.Answer in one or two sentences

16. Define Zwitter ion.
17. Fat deposits in _____ tissues
18. What is mucin?
19. Which amino acid is called imino acid?

20. Give one example for phospholipids
21. Mention the function of hemoglobin
22. Define Iodine number?
23. What is Insulin?
24. What is Edman readent?
25. Mention any two non essential amino acids

SECTION-B (4X5=20)

Answer any FOUR of the following questions:

26. Explain about the structure and function of phospholipids
27. Mention the properties of amino acids
28. Classification of amino acids based on polarity
29. Explain the function of proteins
30. Explain about Ramachandran plot
31. Compare the properties of Globular and Fibrous proteins

SECTION-C (3X10=30)

Answer any THREE of following questions:

32. Explain Classification of Lipids with examples
33. General reaction of amino and carboxyl group in protein
34. Explain about primary,secondary, tertiary and quaternary structure of proteins

35. What is lipoprotein? Mention the types and function of lipoproteins
36. Explain the following
    1. Plasmalogen
    2. R M number
    3. Saponification number
    4. Cerebrosides.